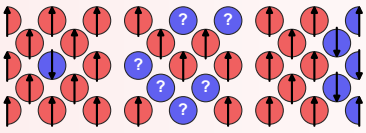
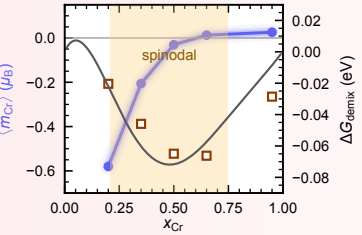
FRUSTRATE → NEST → STIFFEN → DETECT

(a) atoms & moments

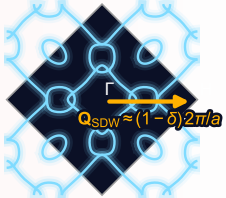
schematic





(b) electrons at E_F

Cr: jack (Γ) nests onto octahedron (H)
⇒ spin- and charge-density waves

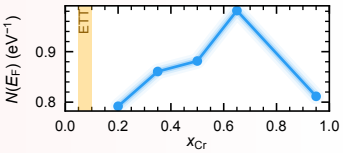


ETT (Lifshitz): adding Cr moves E_F →



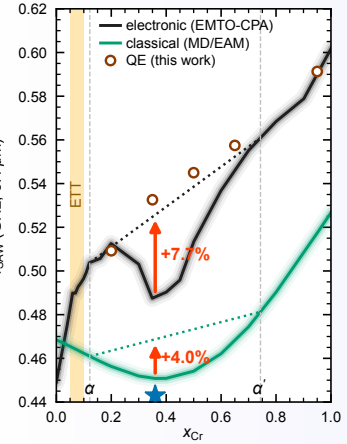
Fe ↑ sheet, 5–10% Cr: √-type cusp in B, C'

schematic



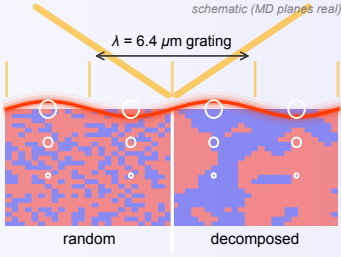
(c) electron → lattice

$C = C_{ion} + C_{band}$, $C_{band} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2\, \delta(\varepsilon_k - E_F)$



(d) surface acoustic waves

schematic (MD planes real)



$C_{ij}\uparrow \Rightarrow G\uparrow \Rightarrow v_R = \eta(\nu)\sqrt{G/\rho}\uparrow \Rightarrow f = v_R/\lambda\uparrow$

MD: 451 → 474 MHz (+5.1%); electronic: +7.7%

10^4 grains, λ = 6.4 μm; QE at 35% Cr

